

PREDICTING FILM REVENUE FROM POSTERS, SYNOPSSES AND TALENT SIGNALS

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Abstract

We built a multimodal pipeline to predict film revenue by combining poster embeddings, synopsis embeddings, and structured metadata to train gradient-boosting models. The largest performance gains came from richer feature engineering, not from model selection.

Background

Pre-release revenue prediction drives studio decisions to greenlight films for production. However, most previous models built for this purpose focus on film metadata (e.g., budget) and talent information (e.g., actor names). That ignores potentially informative data such as the attractiveness of a poster or synopsis. We sought to broaden the scope of analysis by incorporating rich visual and semantic information into our prediction model.

Objective

For pre-release revenue prediction, does multimodal feature engineering matter more than model selection?

Data and Methodology

➤ **3 data sources:** posters + synopsis + metadata

➤ **Datasets:** 

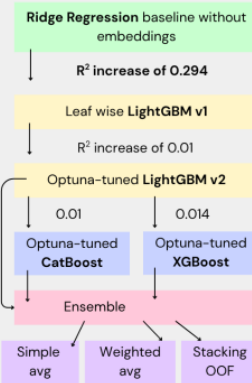
➤ **Preprocessing:** Cut down dataset size from 1.4M to 10k by removing unreleased films (1.37M), films for an adult audience (1.23M), missing budget/revenue values (10k), invalid release dates and empty overviews.

➤ **Train/val/test split:** 70/15/15 temporal split by release date to prevent data leakage.

➤ **Data leakage analysis:** Identified potential post-release information and trained two models with and without access to these columns. If the two models have dramatically different performance, data leakage is likely.

➤ **Feature engineering:** Fusing five modalities of data: film metadata, one-hot encoded genre indicators, synopsis embeddings, poster embeddings, and talent features (expanding median of director and actor prior film revenue). All features scaled on train data only to prevent data leakage.

➤ **Model training flow**

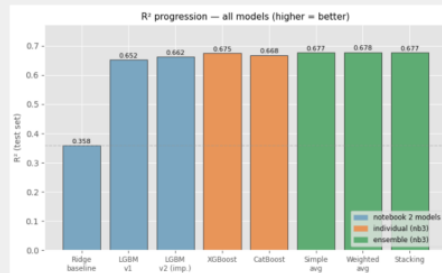


Results and Analysis

$R^2 = 0.677$
final ensemble

+31.9 pp R^2
over baseline

1.71 RMSE
log-revenue units



- Results competitive with published benchmarks (0.50–0.70 R^2). The model explains 67% of variance in revenue, and 30% is due to unobservable factors.
- RMSE means the model's typical error is roughly 5.5× in revenue space, so a film predicted at \$50M could realistically be anywhere from \$9M to \$275M. Best measure of accuracy.
- All 3 ensembling strategies perform nearly identically, likely because the base gradient-boosted tree models share correlated errors, giving the meta-learner little additional signal to exploit.

Implications

SHAP analysis



- As expected, budget is overwhelmingly the dominant predictor.
- Poster embeddings rank highly, but show no directional color structure; the model might be overfitting to noise.
- Director and cast with strong history push predictions up; as does US productions, reflecting Hollywood's advantage.
- Longer runtimes correlate with higher revenue, likely as a proxy for budget.

Recommendation

- Expanding the dataset size and sources
- Adding marketing spend data
- Improving synopsis and poster embeddings
- Adding a dedicated visual branch (CNN)

Conclusion

- The largest improvement in R^2 came from including different modalities in data, rather than model selection or tuning.
- Selecting informative features is crucial for pre-release predictions.
- Different ensemble strategies didn't provide a noticeable difference.

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References: [Dataset IMDb](#) [Dataset TMDB](#)

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